Year 13 Maths - Forces in context

Similar questions - Pulleys, friction and forces at angles

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A rectangular block B rests on a horizontal table. A particle P of mass 2 kg lies on the upper surface of B. P is attached to one end of a light inextensible string which passes over a smooth fixed pulley at the edge of the table; the other end is attached to a particle Q of mass 3 kg hanging freely. The part of the string between P and the pulley is horizontal. The system is released from rest with the string taut. B remains in equilibrium while P slides across it, and the tension in the string is 14.7 N. Take $g = 9.8$.

(a) Find the acceleration of Q. **[2]**

(b) Determine the coefficient of friction between P and B. **[3]**

(c) The coefficient of friction between B and the table is $\frac{1}{7}$. Find the least possible mass of B. **[3]**

2 A block of mass m kg rests on smooth horizontal ground. One end of a light inextensible rope is attached to the block; the rope passes over a smooth pulley and the other end supports a 5 kg object hanging vertically. The part of the rope between the block and the pulley makes an angle of 50° with the horizontal. A force of magnitude X N acts on the block at 20° below the horizontal, directed away from the pulley. The block is in equilibrium. Take $g = 9.8$.

(a) Determine the value of X. **[3]**

(b) The magnitude of the normal contact force of the ground on the block is 147 N. Determine m. **[3]**

Answers

Question 1

(a) For Q: $3g - T = 3a$, so $29.4 - 14.7 = 3a$ and $a = 4.9\ \text{m s}^{-2}$

(b) For P: $T - F = 2a$, so $F = 14.7 - 9.8 = 4.9$ N. $R = 2g = 19.6$, so $\mu = \dfrac{4.9}{19.6} = 0.25$

(c) P exerts a friction force of 4.9 N on B towards the pulley and $R = 19.6$ N downwards. Equilibrium of B: $4.9 \leqslant \frac{1}{7}(9.8M + 19.6)$ so $34.3 \leqslant 9.8M + 19.6$, giving $M \geqslant 1.5$. Least mass = 1.5 kg

Question 2

(a) The rope tension is $T = 5g = 49$ N. Horizontally: $T\cos 50^\circ = X\cos 20^\circ$, so $X = \dfrac{49\cos 50^\circ}{\cos 20^\circ} = 33.5$ N

(b) Vertically: $R + T\sin 50^\circ = mg + X\sin 20^\circ$, so $R = mg + X\sin 20^\circ - T\sin 50^\circ$. With $R = 147$: $mg = 147 - 11.46 + 37.54 = 173.1$, so $m = 17.66$ kg (3 s.f. 17.7)